

COMPANION WORKSHEET TO THE COMBINING ROOM

Appositives in Physics

Syntax as a test of what your physics nouns mean

senior secondary physics

In *The Combining Room*, you practise constructing physics sentences from propositions. In this companion, we slow down over one particular construction: the appositive. The question is no longer only “Can these ideas be combined into a sentence?” but “What does the noun in that sentence commit us to saying exists, acts, varies or represents?”

An **appositive** is a noun or noun phrase placed beside another noun to rename it, classify it, or sharpen its meaning.

Entropy, a state function associated with multiplicity, increases in an isolated system.
appositive

In physics, an appositive can do more than improve style. It forces a question that equations can sometimes conceal:

What kind of thing is this quantity, object, interaction, model or representation?

A good appositive therefore acts as a small test of exactly that question. It can expose whether a student is treating a rate as a substance, a field as a force, a model as an object, or a relation as something possessed by a single body.

1 How tightly to bind it

An appositive earns its place by staying short enough to be said in one breath and settled enough not to need arguing for. That is what separates it from a footnote. A footnote is where a classification gets qualified, contested, or given a second sense worth distinguishing from the first; an appositive is where a classification is simply supplied, cleanly enough that the sentence can keep moving.

Comma	Entropy, a state function associated with multiplicity, increases in an isolated system.
Dash	Entropy—not a force, not a fluid, but a state function—increases in an isolated system.
Footnote	Entropy increases in an isolated system. ¹

The comma version is the default and should stay the default. The dash version is for the rare case where the classification is itself the point of the sentence, the thing the reader’s eye should catch, rather than a courtesy supplied in passing; used often it stops signalling anything. The footnote is for content that has stopped classifying and started arguing, qualifying, or distinguishing senses, which is a different job and deserves the extra room.

¹The claim needs a footnote once it stops classifying and starts arguing: whether entropy is best read realistically, as a property the system has, or epistemically, as a measure of what an observer does not know about it, is a live question in the philosophy of statistical mechanics and does not settle inside one sentence.

For each item, decide whether the classifying phrase belongs as a comma appositive, a dash appositive, or ought to be moved to a footnote. Give one reason.

1. “Momentum, a vector quantity equal to mass multiplied by velocity, is conserved in an isolated system.”

Register: _____ Reason:

2. “A virtual photon is not a particle in the ordinary sense but an internal element of a perturbative calculation, and whether such elements should be read as representing anything physical at all, rather than as bookkeeping devices internal to a particular calculational scheme, is a question the diagrams themselves cannot answer.”

Register: _____ Reason:

3. “Density remains constant when a homogeneous sample is divided, because it is an intensive ratio of mass to volume rather than an extensive amount.”

Register: _____ Reason:

2 Warm-up: what kind of thing is it?

For each term, write a short appositive that classifies it before saying what it does.

Example

Acceleration, _____, may result from a change in speed, direction, or both.

Possible completion: the rate of change of velocity with time.

1. **Momentum**, _____, is conserved in an isolated system.
2. **Current**, _____, is measured in amperes.
3. **Potential difference**, _____, is measured between two points.
4. **Temperature**, _____, helps determine the direction of spontaneous thermal energy transfer.
5. **A ray**, _____, is drawn perpendicular to the wavefronts.

6. **Gravitational field strength**, _____, has units of N kg^{-1} .
7. **Half-life**, _____, is independent of the initial number of unstable nuclei.
8. **Proper time**, _____, is measured by a clock present at both events.

3 Choose the appositive that preserves the physics

Circle the best completion. Then explain what is wrong with one rejected option.

1. Mechanics

Force, . . . , can change the momentum of a system.

- A. the energy stored in a moving object
- B. an interaction between systems
- C. the tendency of an object to keep moving

Rejected option: _____
Why it fails physically:

2. Thermal physics

Entropy, . . . , increases in an isolated system undergoing an irreversible process.

- A. the amount of disorder inside a system
- B. a force driving systems towards equilibrium
- C. a state function associated with the multiplicity of accessible microstates

Rejected option: _____
Why it fails physically:

3. Waves

A ray, . . . , is used to represent the direction of propagation.

- A. a narrow material thread of light
- B. the trajectory followed by a single photon
- C. a geometrical construction used in a model

Rejected option: _____

Why it fails physically:

4. Quantum physics

The wavefunction, . . . , evolves according to the Schrödinger equation.

- A. a mathematical object used to calculate probabilities for measurement outcomes
- B. a physical ripple distributed through space
- C. the cloud of charge surrounding every nucleus

Rejected option: _____

Why it fails physically:

5. Fields

Electric potential, . . . , is defined at a location in an electric field.

- A. the electrical energy stored at a point
- B. the potential energy per unit charge relative to a chosen reference
- C. the force possessed by a charge at that point

Rejected option: _____

Why it fails physically:

You have now asked, several times over, what kind of thing a physics noun is claiming to be. That question has a name: philosophers call it an **ontological** question, and asking it, of a noun, is asking what kind of thing we are claiming it denotes. Naming the question does not change anything you have been doing. It only gives you a word for it, so the sections that follow can use it.

4 Repair the ontology

Each sentence is grammatically plausible and physically dangerous. Rewrite it with an appositive that makes the category explicit.

1. “**The object’s inertia keeps it moving.**”

Rewrite:

2. “**Current is the electricity flowing through the wire.**”

Rewrite:

3. “**Heat, the energy inside a hot object, flows into the colder object.**”

Rewrite:

4. “**Centripetal force, the extra inward force, keeps the satellite in orbit.**”

Rewrite:

5. “**Voltage, the electrical push supplied by the battery, is used by the resistor.**”

Rewrite:

6. “**A field line, the path followed by a test particle, shows where it will move.**”

Rewrite:

5 Appositives inside full explanations

Now overleaf combine each set of propositions into *one* scientifically defensible sentence. Your sentence must include at least one appositive.

A. Circular motion

- A body moves at constant speed.
- Its direction changes continuously.
- Its velocity therefore changes.
- A resultant force acts towards the centre.
- Centripetal acceleration is the inward acceleration associated with this changing velocity.

Your sentence:

B. Ohmic behaviour

- The potential difference across a resistor increases.
- Its temperature remains constant.
- The current increases in proportion.
- Resistance is the ratio V/I under specified conditions.

Your sentence:

C. Thermal equilibrium

- Two bodies are placed in thermal contact.
- Energy can be exchanged between them.
- Net energy transfer ceases when equilibrium is reached.
- Temperature is a state variable associated with thermal equilibrium.

Your sentence:

Then answer: what has your appositive committed you to saying temperature *is*?

D. Refraction

- A wave crosses a boundary.
- Its speed changes.
- Its frequency remains fixed by the source.
- Its wavelength changes.
- Refractive index is a dimensionless ratio relating wave speed in vacuum to wave speed in the medium.

Your sentence:

E. Radioactive decay

- Each unstable nucleus has a fixed probability of decay per unit time.
- The nuclei do not age.
- The activity is proportional to the number of undecayed nuclei.
- Half-life is a statistical timescale for the expected population to halve.

Your sentence:

6 Which appositive is doing useful work?

Read each pair. Tick the more useful sentence for physics and state why.

Pair 1

- ☐ Density, *the amount of matter packed into a volume*, remains constant when a homogeneous sample is divided.
- ☐ Density, *an intensive ratio of mass to volume*, remains constant when a homogeneous sample is divided.

Why?

Pair 2

- ☐ Resistance, *a force opposing electrons*, increases as a filament gets hotter.
 - ☐ Resistance, *the ratio of potential difference to current under specified conditions*, increases as a filament gets hotter.
- Why?

Pair 3

- ☐ Magnetic flux, *the amount of magnetism passing through a loop*, depends on area and orientation.
 - ☐ Magnetic flux, *a scalar measure of magnetic field through a surface*, depends on area and orientation.
- Why?

Pair 4

- ☐ Uncertainty, *the error in a measurement*, should be made as small as possible.
 - ☐ Uncertainty, *a reported range expressing plausible values of a measured quantity*, should be justified from the measurement process.
- Why?

7 From appositive to equation

An appositive can sometimes anticipate the mathematical structure of a quantity. Complete the sentence first; then write the equation it suggests.

Quantity	Appositive	Equation
Momentum	a vector quantity equal to mass multiplied by velocity	
Current	the rate at which charge crosses a section	
Pressure	force per unit area acting normal to a surface	
Power	the rate of energy transfer	
Electric field strength	force per unit positive test charge at a location	
Gravitational potential	potential energy per unit mass relative to a reference	

8 Make the appositive earn its place

For each sentence below, decide whether the appositive is:

- defining;
- specifying a condition;
- marking epistemic status;
- classifying;
- marking a model;
- unnecessary.

1. The ideal gas, *a model in which particle volume and intermolecular forces are neglected*, obeys $pV = nRT$ within an appropriate regime.

Type: _____

2. A virtual photon, *an internal element of a perturbative calculation*, appears in the interaction diagram.

Type: _____

3. Proper time, *the interval between events occurring at the same place in one inertial frame*, is invariant.

Type: _____

4. Power, *measured in watts*, is the rate of energy transfer.

Type: _____

5. Terminal speed, *the constant speed reached when the resultant force becomes zero*, depends on the interaction between the moving body and the surrounding medium.

Type: _____

9 Extension: improve a short physics paragraph

The paragraph below contains several nouns whose ontology is vague.

A satellite is kept in orbit by centripetal force. Gravity pulls it towards the Earth, while inertia keeps it moving forwards. The field gives the satellite potential energy. If the satellite moves farther away, gravity becomes weaker and the potential energy becomes less negative.

Rewrite the paragraph. Use at least **three appositives**. Your revision should distinguish:

- interaction from motion;
- force from acceleration;
- field from field line or force;
- potential from potential energy;
- object property from system property.

10 Reflection

Choose one term from this worksheet that you thought you understood before beginning.

Term: _____

Complete:

1. Before the worksheet I tended to treat it as ...
2. The appositive I would now attach to it is ...
3. This changes the physics because ...

Teacher notes / possible responses

These are models rather than unique answers.

Model answer

Force: an interaction between systems.

Momentum: a vector quantity equal to mass multiplied by velocity.

Current: the rate of flow of charge through a chosen cross-section.

Potential difference: the energy transferred per unit charge between two points.

Temperature: a state variable associated with thermal equilibrium.

Ray: a geometrical construction showing the direction of propagation or energy transfer.

Gravitational field strength: gravitational force per unit mass at a location.

Half-life: the time required for the expected number of undecayed nuclei to halve.

Proper time: the interval between events occurring at the same place in one frame.

Model answer

Possible mechanics repair:

Centripetal acceleration, *the inward acceleration associated with a continuously changing velocity direction*, requires a resultant inward force; for an orbiting satellite this force is supplied by the gravitational interaction between the Earth and the satellite.

Model answer

Possible thermal-equilibrium sentence:

Temperature, *a state variable associated with thermal equilibrium*, becomes equal for two bodies in thermal contact when net energy transfer between them ceases.

Model answer

Possible refraction sentence:

Refractive index, *the dimensionless ratio c/v for a wave in a medium*, changes across the boundary, so the wave speed and wavelength change while the source-determined frequency remains constant.

Model answer

Possible paragraph repair:

The Earth exerts a gravitational force on the satellite, producing centripetal acceleration, *the inward acceleration required for continuously changing velocity*. The satellite's inertia, *its resistance to changes in velocity rather than a force*, does not push it forwards. The gravitational field, *a description assigning gravitational force per unit mass to locations around the Earth*, allows us to define gravitational potential. As the satellite moves farther away, the gravitational interaction weakens and the gravitational potential energy of the Earth–satellite system increases, becoming less negative relative to the chosen zero at infinity.

Model answer

Register task, model answers:

1. Comma. The classification is short, uncontested, and needed at once for the sentence to make sense.
2. Footnote, and the item is written at the length a footnote should be, not a dash, to make the point concretely: once a classification starts hedging its own status, it has left appositive territory.
3. Comma, though a fluent student may prefer to fold it into a because-clause as shown, which is the sentence-combining skill from earlier sections meeting this one.

Teaching move. Ask students to read their appositive aloud while suppressing the surrounding sentence. If the phrase on its own sounds like a substance, a hidden agent, a possession, or a diagram mistaken for a physical object, the ontology probably needs repairing.

Adapted as a printable companion to the uploaded *The Combining Room: Syntax as Ontology in School Physics*. The worksheet retains its use of appositives, sentence combining, agent restoration and model-status language, while concentrating the tasks at 16–18 level.